

Education

2023.02 – Present	Ph.D. candidate of Mechanical Engineering, Robotics Research Group & Processing Speech and Images, KU Leuven . Supervisor: Prof. Renaud Detry	
2019.09 – 2022.07	Master of Computer Science, Key Laboratory of Machine Perception, Peking University . Supervisor: Prof. Hong Liu	GPA: 3.6/4
2015.09 - 2019.07	Bachelor of Mechanical Engineering, School of Mechanical & Automotive Engineering, South China University of Technology (SCUT)	GPA: 3.74/4

Experience

Neurobotics

KU Leuven

Ph.D. candidate, Robotics Research Group & Processing Speech and Images

2022.03 – Present

- Summary: To enable monkeys with invasive BMI to navigate and manipulate with robot
 - Develop a shared control system for brain-machine interface (BMI).
 - Validate the system on monkeys with invasive BMIs, enabling them to perform navigation in a simulated cluttered indoor Unity environment.

PlatoOCR

Bytedance, Shanghai

Developer, Data-EDU Group

2021.07 – 2021.10

- Summary: To detect the paragraphs and characters end-to-end
 - Design the multi-tasking model for paragraph segmentation and character detection.
 - Develop a weekly supervise training paradigm for limited character annotations.
 - Achieve better performance on real-scene data.

Smart Shop System

Peking University

Leader, Goods Detection Group

2020.07 – 2022.07

- Summary: To detect whether the goods are purchased.
 - Design the auto-labeling system and building the goods dataset.
 - Develop a smart retail shelf with in-shelf cameras for goods detection.
 - Design detection algorithm by using deep learning methods, achieving a 98% detection rate.

Object Detection for Underwater Robot

Pengcheng National Laboratory, Shenzhen

Developer, Underwater Image Processing Group

2019.09 – 2021.06

- Summary: Developing Underwater Object Detection Algorithm for Underwater Robot.
 - Realize detection of marine products in underwater environment by deep learning.
 - Realize robustness of detection algorithm in different underwater environments.

Academic Achievement

Total Publication: 23, Google Scholar citation: 1558

Representative Publications

- **Song, P.**, Li, P., Dai, L., Wang, T. and Chen, Z., 2023. Boosting R-CNN: Reweighting R-CNN samples by RPN’s error for underwater object detection. *Neurocomputing*, 530, pp.150-164. (Citation: 305)
- **Song, P.**, Li, P., Aertbeliën, E. and Detry, R., 2024, May. Robot trajectron: Trajectory prediction-based shared control for robot manipulation. *ICRA 2024*. (Citation: 23)
- **Song, P.**, Li, P. and Detry, R., 2024, September. Implicit grasp diffusion: Bridging the gap between dense prediction and sampling-based grasping. *CORL 2024*.
- **Song, P.**, Du, Y., Saussus, O., De Schrijver, S., Caprara, I., Janssen, P. and Detry, R., 2025. Robot Trajectron V2: A Probabilistic Shared Control Framework for Navigation. **Submitted to IJRR (Major Revision)**.
 - 2025 IROS SASA workshop Best Poster Award 3rd place
- **Song, P.**, Hu, Y., Li, P. and Detry, R., 2025. Equivariant Volumetric Grasping. **Submitted to IJRR**.
- Du, Y., **Song, P.**, Hu, Y. and Detry, R., 2026. ELVIS: Ensemble-Calibrated Latent Imagination for Long-Horizon Visual MPC. *RSS 2026*.
- Hu, Y., **Song, P.**, Wen, K. and Detry, R., 2025. Mini Diffuser: Fast Multi-Task Diffusion Policy Training Using Two-Level Mini-Batches. *RAL 2026*.

Prize

- 2025 IROS SASA workshop Best Poster Award 3rd place
- 2020 Scientific Research Progress Award (Peking University)
- Silver Prize of 2018 ”Challenge Cup” National Entrepreneur Contest for Undergraduate Student
- 2018 Second Class School Scholarship; 2017 Third Class School Scholarship (SCUT)
- 2016 “xiaochu” Enterprise Scholarship (SCUT, Top 1%)

Skills

Theory	Machine Learning, Robotics, Deep Learning, Computer Vision, Reinforcement Learning
Programming	python, ROS2, C/C++, MATLAB, pytorch, keras, opencv
Software	Pybullet, Mujoco, IssacSim, Office, PS, CAD
Language	English (IELTS: 7), Japanese (JPLT N1), Dutch (A2)

Activity

- Work as a volunteer in Leuven AI conference (600 years KU Leuven – 70 years AI)